Unit 2 - Introduction to Devops

1) Define Devops

Devops is a culture, practice, and set of tools that brings Developement (Dev) and operations (ops) teams together to deliver software faster, more reliably and with better quality.

2) What is DevOps ?

Devops is a collaboration approach that:
- improve communication between developers and operations.
- Automates software developement, testing, and ~~develope~~ deployment
- Ensure faster delivery with fewer error.
- focuses on continous improvement.

Devops = Developement + operations + Automation + continous delivery

3) SDLC Models

Common models.

1) Waterfall Model
- linear, step-by-step
- Each phase must finish before the next
- slow and rigid.

2) Agile Model
- iterative and increamental.
- flexible, frequent delivery
- customer feedback based

3) Spiral Model
- combines design + prototyping
- Risk analysis is important

4) V Model
- Verification (static, early stage
  Validation (dynamic, later stage
- Testing is parallel to development

5) DevOps Model
- Automation, CI/CD, quick releases
- Integrates development + operations

4) Lean
- Focuses on reducing waste
- improves productivity by removing unnecessary tasks
- principles : value, flow, pull, perfection

5) ITIL (information Technology infrastructure Library)
- A framework to manage I.T services
- Ensures stability, quality and effeciency
- covers service strategy, design, transition, operation, improvement

6) Agile
- software developement methodology
- Uses sprints (2-4 weeks)
- customer involvement
- Quick delivery
- flexible to change

7) Why DevOps ?
DevOps is needed because it :
- Removes the gap between development and operation
- Reduces deployment failures
- improves speed, quality, and stability
- Automates repetitive tasks
- provides continuous delivery
- Enables quick bug fixing

8) History of DevOps

- 2007-2008 : patrick Debois introduced the concept.
- 2009 : First "DevOpsDays" conference in Belgium
- Grew due to the need for faster software delivery
- Became standard in cloud-based companies (Amazon, google, Netflix)

9) DevOps stakeholder

People involved in DevOps :

- Developer
- Tester
- operations / IT admins
- Business analysis
- project managers
- product owners
- security professionals (DevSecOps)
- customers / end users

10) DevOps Goals

- faster software delivery
- High quality and fewer bugs
- Automation of pipeline
- Better collaboration
- continuous integration & continuous deployment (CI/CD)
- faster recovery from failure
- improvement in customer satisfaction.

1) Important DevOps terminology.

key terms

1) CI (Continuous Integration)

CI means developers frequently merge code into a shared repository, where automated builds and tests run to detect errors early.

2) CD (continuous Delivery / continuous Deployment)

CD automates the process of delivering or deploying code to production, ensuring fast and reliable releases.

3) Pipeline

A pipeline is a sequence of automated steps like build → test → deploy that ensures smooth s/w delivery

4) Version control

Version control stores and manages changes to code, enabling teamwork, tracking, and roll back (e.g. Git)

5) Artifact

An Arbifact is the final built output of a project such as a .jar, .war, binary or Docker image generated during the build process.

6) Infrastructure as code (IaC)

IaC means managing and provisioning infrastructure (servers, networks, etc) through code instead of manual setup.

7) Containerization

Containerization packages application with all dependancies into lightweight containers (e.g Docker) to run consistently anywhere.

8) Monitoring

Monitoring continuously tracks the performance, health, and errors of application and system to detect issue early.
(tracking health of system)

12) DevOps Perspective

DevOps focuses on:

- collaboration
- Automation
- Integration
- Monitoring
- continuous improvement
- faster feedback loops
- shared responsibility

13) DevOps and Agile

Agile

- focuses on developement
- Works in small iteration
- frequent customer feedback

DevOps

- Extends Agile by including operations.
- Helps in deploy what agile builds
- Automates testing & delivery

Agile + DevOps = faster, high-quality s/w delivery

14) DevOps Tools

DevOps uses many tools for automatic

- version control
  Git, GitHub, Git Lab, Bit bucket
- Build Tools
  Maven, Gradle
- CI/CD Tools
  Jenkins, GitHub Actions, Gitlab CI, Circle CI
- Configuration Management
  Ansible, puppet, chef
- Containerization
  Docker, Podman

- Orchestration
  Kubernetes, Docker swarm
- Monitoring Tools
  Prometheus, Grafana, Nagios
- Cloud platform
  AWS, Azure, GCP

15) ~~Conf~~ Continuous Integration (CI)
CI Means:
- Developers upload code frequently
- code is automatically built and tested
- Detect errors early
- Ensures stable codebase

Tools: Jenkins, GitLab CI, GitHub actions

16) Continuous Deployment (CD)
CD Means:
- fully automated release
- code moves from developement → testing → production automatically
- No manual interventions
- faster delivery to customers

Tools -: Jenkins, Spinnaker, Argo CD.